

# Morphology-aware Joint-space DS Transform for Robust Action Transfer

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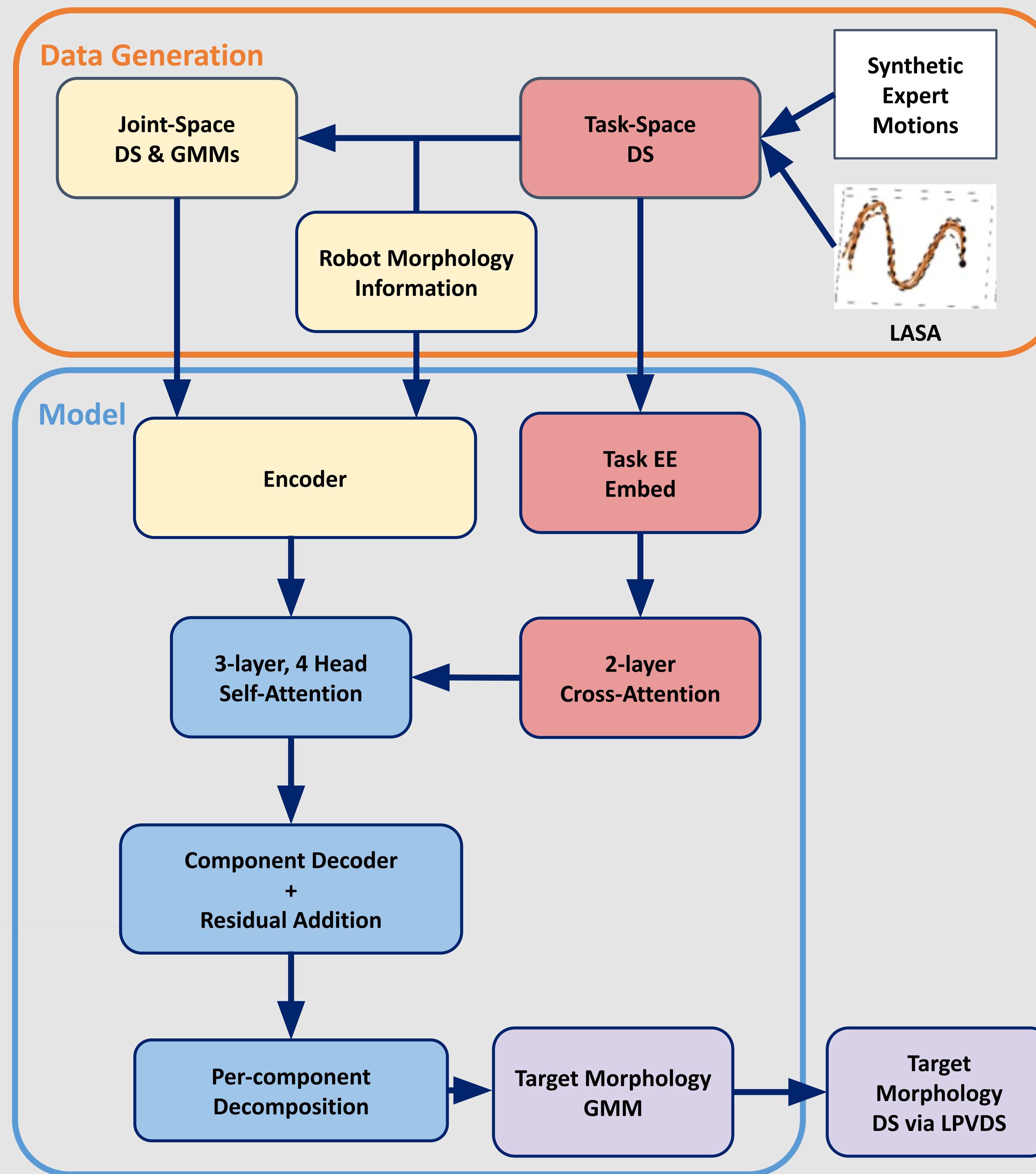
## INTRODUCTION & MOTIVATION

- **Task-space** allows workspace motions to be generalised between robots.
- However, they are only descriptive of an **end-effector**, with control of each joint fully at the discretion of a low-level controller.
- If we wish to **control joints** and enforce constraints, we must operate in **joint-space** DS.
- Our approach enables a joint-space DS to be directly **one-shot transformed** between robots.
- This allows significant adaptability in dynamic environments, such as manufacturing.
- It enables **different robots** to operate cohesively and be **interchanged** upon short notice maintenance or downtime.
- We simulate and test on robot arms at **7 DoF**.

## METHODS

1. Sample expert trajectories from task-space DS.
2. Generate joint-space DS for various robots with varying link-lengths at of morphology.
3. Show proof-of-concept for 2 and 3 DoF robots.
4. Train more advanced model on overactuated Kuka and Franka Panda robots in 3D.
5. Validate conformity to joint constraints, reproducibility of complex trajectories, and adaptability to unseen robots or morphologies.

## ML Model Architecture and Training Pipeline



## RESULTS

| Model                      | FK(Mu)          | ± Std        | vs Naive      | FK(Mu) lg-gap   | vs Naive      | Δ Oracle      |
|----------------------------|-----------------|--------------|---------------|-----------------|---------------|---------------|
| Naive (no transfer)        | 0.0859 m        | —            | —             | 0.1323 m        | —             | 63.6 mm       |
| MLP (5-scale)              | 0.0791 m        | 0.032        | +7.9%         | 0.0810 m        | +38.8%        | 56.8 mm       |
| Transformer (5-scale)      | 0.0478 m        | 0.016        | +44.4%        | 0.0496 m        | +62.5%        | 25.5 mm       |
| MLP (11-scale)             | 0.0566 m        | 0.018        | +34.1%        | 0.0580 m        | +56.1%        | 34.3 mm       |
| Transformer (11-scale)     | 0.0395 m        | 0.012        | +54.0%        | 0.0411 m        | +68.9%        | 17.2 mm       |
| Trans V3 (dropout+traj)    | 0.0360 m        | 0.012        | +58.1%        | 0.0369 m        | +72.1%        | 13.7 mm       |
| Trans V4 (V2 arch)         | 0.0322 m        | 0.013        | +62.5%        | 0.0344 m        | +74.0%        | 9.9 mm        |
| <b>Trans V5 (V3+xattn)</b> | <b>0.0275 m</b> | <b>0.011</b> | <b>+68.0%</b> | <b>0.0302 m</b> | <b>+77.2%</b> | <b>5.2 mm</b> |
| Any model + TTO            | <b>0.002 m</b>  | <b>0.002</b> | <b>+97+%</b>  | <b>0.002 m</b>  | <b>+98+%</b>  | -20 mm        |
| Oracle (GT target GMM)     | 0.0223 m        | —            | —             | —               | —             | 0 mm          |

| Generalization                                                                    | FK(Mu)  | vs Naive | Note                       |
|-----------------------------------------------------------------------------------|---------|----------|----------------------------|
| <i>Interpolation</i> (held-out scales 0.6, 0.8, 1.2, 1.4 — within training range) |         |          |                            |
| Trans V4 (held-out)                                                               | 0.036 m | +58%     | only 0.3 mm gap vs trained |
| <i>Extrapolation</i> (scales 0.35–1.65 — outside training range)                  |         |          |                            |
| Trans V4                                                                          | 0.036 m | +74%     | best extrapolation         |
| Trans V3                                                                          | 0.041 m | +71%     | —                          |
| <i>With TTO</i> (any model)                                                       |         |          |                            |
| Any + TTO                                                                         | 0.002 m | +97+%    | dominates all conditions   |

### Key

**FK(Mu):** Mean nearest-neighbor EE distance from predicted GMM means to task-space reference ( $\text{FK}(\mu_k) \rightarrow \mathbf{p}^{\text{task}}$ ). Lower is better.  
**lg-gap:** Pairs with scale difference  $|s_{\text{src}} - s_{\text{tgt}}| \geq 0.4$  (harder transfers).  
**Δ Oracle:** Gap between model FK(Mu) and oracle FK(Mu) = 22.3 mm.  
**TTO:** Test-Time FK Refinement — 100 Adam steps optimizing  $\hat{\theta}$  via differentiable FK. Warm-started by model prediction. Brings any model to ~2 mm (beats oracle).  
**Dataset:** diverse\_v2, 11 scales (0.5–1.5), 50 batches, all ordered pairs.

1. Wins: GMMTransformer + residuals + task attention → ~79→~5 mm; better data +  $\mu$ -weight + dropout help.
2. Best boost: Chamfer FK + test-time FK optimization → ~2 mm (dominant gain).
3. Fails: Flow matching, flat MLP, bad FK loss, augmentation, ensembles.

